Online on xv6 System call

Section: B2 — **Time:** 30 minutes

In this assignment, you will implement a per-process stopwatch using xv6's global kernel ticks timer. This allows a process to measure elapsed kernel time between two events.

Required System Calls:

1. `startStopwatch()` : Records the current kernel ticks counter into a new field in the calling process's `struct proc`. Returns 0 on success.
2. `readStopwatch()` : Calculates and returns `ticks - start_ticks` (elapsed ticks since `startStopwatch` was called). Returns -1 if the stopwatch was never started by this process.

Mechanics:

- Add a `start_ticks` field and an `is_running` boolean/flag field to `struct proc` in `kernel/proc.h`.
- Access the global `ticks` variable (defined in `kernel/trap.c`).

Required User Commands:

3. `watch_start`
4. `watch_read`

See the sample I/O for clarification:

```
$ watch_read
Stopwatch not started!
$ watch_start
Stopwatch started
$ sleep 10
$ watch_read
Elapsed time: 10 ticks
$ sleep 5
$ watch_read
Elapsed time: 15 ticks
```

Submission:

```
git add --all
git diff HEAD > ../{studentID}.patch
```